

Question #1 of 37

Question ID: 1473935

Diversification can reduce:

- A) unsystematic risk.
- B) macroeconomic risks.
- C) systematic risk.



Explanation

Systematic risk reflects factors that have a general effect on the security markets as a whole, and cannot be diversified away. Macroeconomic risk comes in many forms, and it is usually considered systematic risk. Unsystematic risk can be reduced through diversification.

(Module 36.1, LOS 36.a)

Question #2 of 37

Question ID: 1473926

Which of the following is NOT an assumption necessary to derive the arbitrage pricing theory (APT)?

- A) The priced factors risks can be hedged without taking short positions in any portfolios.
- B) Asset returns are described by a k-factor model.
- C) A large number of assets are available to investors.



Explanation

APT does not prohibit short positions.

(Module 36.1, LOS 36.a)

Question #3 of 37

Question ID: 1473947

Assume you are considering forming a common stock portfolio consisting of 25% Stonebrook Corporation (Stone) and 75% Rockway Corporation (Rock). As expressed in the two-factor returns models presented below, both of these stocks' returns are affected by two common factors: surprises in interest rates and surprises in the unemployment rate.

$$R_{\text{Stone}} = 0.11 + 1.0F_{\text{Int}} + 1.2F_{\text{Un}} + \epsilon_{\text{Stone}}$$

$$R_{\text{Rock}} = 0.13 + 0.8F_{\text{Int}} + 3.5F_{\text{Un}} + \epsilon_{\text{Rock}}$$

Assume that at the beginning of the year, interest rates were expected to be 5.1% and unemployment was expected to be 6.8%. Further, assume that at the end of the year, interest rates were actually 5.3%, the actual unemployment rate was 7.2%, and there were no company-specific surprises in returns. This information is summarized in Table 1 below:

Table 1: Expected versus Actual Interest Rates and Unemployment Rates

	Actual	Expected	Company-specific returns surprises
Interest Rate	0.053	0.051	0.0
Unemployment Rate	0.072	0.068	0.0

What is the predicted return for Stonebrook if the return unexplained by the model was -1%?

A) 12.00%



B) 10.68%.



C) 1.40%.



Explanation

The actual return uses the unemployment and interest rate surprises as follows:

The returns for a stock that are correlated with surprises in interest rates and unemployment rates can be expressed using a two-factor model as:

$$R_i = a_i + b_{i,1}F_{Int} + b_{i,2}F_{Un} + \varepsilon_i$$

where:

R_i = the return on stock i

a_i = the expected return on stock i

$b_{i,1}$ = the factor sensitivity of stock i to unexpected changes in interest rates

F_{Int} = unexpected changes in interest rates (the interest factor) = $.053 - .051 = .002$

$b_{i,2}$ = the factor sensitivity of stock i to unexpected changes in the unemployment rate

F_{Un} = unexpected changes in the unemployment rate (the unemployment rate factor) = $.072 - .068 = .004$

ε_i = a mean-zero error term that represents the part of asset i 's return not explained by the two factors.

Thus the actual return is: $0.11 + (1.0)(0.002) + (1.2)(0.004) - 0.01 = 0.1068$ or 10.68%

(Module 36.2, LOS 36.d)

Marcie Deiner is an investment manager with G&G Investment Corporation. She works with a variety of clients who differ in terms of experience, risk aversion and wealth. Deiner recently attended a seminar on multifactor analysis. Among other things, the seminar taught how the assumptions concerning the Arbitrage Pricing Theory (APT) model are different from those of the Capital Asset Pricing Model (CAPM). One of the examples used in the seminar is below.

$$E(R_i) = R_f + f_1 B_{i,1} + f_2 B_{i,2} + f_3 B_{i,3} \text{ where: } f_1 = 3.0\%, f_2 = -40.0\%, \text{ and } f_3 = 50.0\%.$$

Beta estimates for Growth and Value funds for a three factor model			
	Factor 1	Factor 2	Factor 3
Betas for Growth	0.5	0.7	1.2
Betas for Value	0.2	1.8	0.6

For the model used as an example in the seminar, if the T-bill rate is 3.5%, what are the expected returns for the Growth and Value Funds?

$$E(R_{\text{Growth}}) \quad E(R_{\text{Value}})$$

- A)** 3.1% -3.16% 
- B)** 33.5% -41.4% 
- C)** 37.0% -37.9% 

Explanation

$E(R_{\text{Growth}}) = 0.035 + 0.03(0.5) - 0.4(0.7) + 0.5(1.2) = 0.035 + 0.015 - 0.28 + 0.6 = 0.37$ or 37.0%

$E(R_{\text{Value}}) = 0.035 + 0.03(0.2) - 0.4(1.8) + 0.5(0.6) = 0.035 + 0.006 - 0.72 + 0.30 = -0.379$ or -37.9%

(Module 36.1, LOS 36.a)

Question #5 of 37

Question ID: 1473930

The Arbitrage Pricing Theory (APT) has all of the following characteristics *EXCEPT* it:

- A)** assumes that asset returns are described by a factor model. 
- B)** is an equilibrium pricing model. 

C) assumes that arbitrage opportunities are available to investors.



Explanation

The APT assumes that *no* arbitrage opportunities are available to investors.

(Module 36.1, LOS 36.a)

Question #6 of 37

Question ID: 1473928

Which of the following is not an assumption of the arbitrage pricing theory (APT)?

A) The market contains enough stocks so that unsystematic risk can be diversified away.



B) Returns on assets can be described by a multi-factor process.



C) Security returns are normally distributed.



Explanation

APT does not require that security returns be normally distributed.

(Module 36.1, LOS 36.a)

Question #7 of 37

Question ID: 1473950

A tracking portfolio is a portfolio with:

A) factor sensitivities of zero to all factors, positive expected net cash flow, and an initial investment of zero.



B) a factor sensitivity of one to a particular factor in a multi-factor model and zero to all other factors.



C) a specific set of factor sensitivities designed to replicate the factor exposures of a benchmark index.



Explanation

A tracking portfolio is a portfolio with a specific set of factor sensitivities designed to replicate the factor exposures of a benchmark index. A factor portfolio is a portfolio with a factor sensitivity of one to a particular factor and zero to all other factors. An arbitrage portfolio is a portfolio with factor sensitivities of zero to all factors, positive expected net cash flow, and an initial investment of zero.

(Module 36.3, LOS 36.f)

Question #8 of 37

Question ID: 1473939

Assume you are attempting to estimate the equilibrium expected return for a portfolio using a two-factor arbitrage pricing theory (APT) model. Assume that you have estimated the risk premium for factor 1 to be 0.02, and the risk premium for factor 2 to be 0.03. The sensitivity of the portfolio to factor 1 is -1.2 and the portfolio's sensitivity to factor 2 is 0.80 . Given a risk free rate equal to 0.03 , what is the expected return for the asset?

A) 5.0%.



B) 3.0%.



C) 2.4%.

**Explanation**

The general form of the two-factor APT model is: $E(R_{\text{Port}}) = R_F = \lambda_1\beta_1 + \lambda_2\beta_2$, where the λ 's are the factor risk premiums and the β 's are the portfolio's factor sensitivities. Substituting the appropriate values, we have:

$$R_{\text{Port}} = 0.03 + 0.02(-1.2) + 0.03(0.80) = 3.0\%$$

(Module 36.1, LOS 36.c)

Question #9 of 37

Question ID: 1473953

Janice Barefoot, CFA, has been managing a portfolio for a client who has asked Barefoot to use the Dow Jones Industrial Average (DJIA) as a benchmark. In her first year Barefoot managed the portfolio by choosing 29 of the 30 DJIA stocks. She selected a non-DJIA stock in the same industry as the omitted stock to replace that stock. Compared to the DJIA, Barefoot has placed a higher weight on the financial stocks and a lower weight on the other stocks still in the portfolio. Over that year, the non-DJIA stock in the portfolio had a negative return while the omitted DJIA stock had a positive return. The portfolio managed by Barefoot outperformed the DJIA. Based on this we can say that the return from factor tilts and asset selection were:

A) both positive.



B) positive and negative respectively.



C) negative and positive respectively.

**Explanation**

Since the replacement of the asset obviously had a negative effect, the tilting towards financial stocks must have been positive to not only compensate for the loss but produce a portfolio return greater than the DJIA.

(Module 36.3, LOS 36.f)

Question #10 of 37

Question ID: 1473943

Identify the *most* accurate statement regarding multifactor models from among the following.

Macroeconomic factor models include explanatory variables such as real GDP growth and the price-to-earnings ratio and fundamental factor models include explanatory variables such as firm size and unexpected inflation. 

Macroeconomic factor models include explanatory variables such as the business cycle, interest rates, and inflation, and fundamental factor models include explanatory variables such as firm size and the price-to-earnings ratio. 

Macroeconomic factor models include explanatory variables such as firm size and the price-to-earnings ratio and fundamental factor models include explanatory variables such as real GDP growth and unexpected inflation. 

Explanation

Macroeconomic factor models include multiple risk factors such as the business cycle, interest rates, and inflation. Fundamental factor models include specific characteristics of the securities themselves such as firm size and the price-to-earnings ratio.

(Module 36.2, LOS 36.d)

Question #11 of 37

Question ID: 1473931

Which of the following does NOT describe the arbitrage pricing theory (APT)?

A) It is an equilibrium-pricing model like the CAPM. 

B) It requires a weaker set of assumptions than the CAPM to derive. 

C) There are assumed to be at least five factors that explain asset returns. 

Explanation

APT is a k-factor model, in which the number of factors, k, is assumed to be a lot smaller than the number of assets; no specific number of factors is assumed.

(Module 36.1, LOS 36.a)

Question #12 of 37

Question ID: 1473932

One of the assumptions of the arbitrage pricing theory (APT) is that there are no arbitrage opportunities available. An arbitrage opportunity is:

- A) a factor portfolio with a positive expected risk premium. 
- B) an investment that has an expected positive net cash flow but requires no initial investment. 
- C) a portfolio with factor exposures that sum to one. 

Explanation

One of the three assumptions of the APT is that there are no arbitrage opportunities available to investors among these well-diversified portfolios. An arbitrage opportunity is an investment that has an expected positive net cash flow but requires no initial investment.

All factor portfolios will have positive risk premiums equal to the factor price for that factor. An arbitrage opportunity does not necessarily require a return equal to the risk-free rate, and the factor exposures for an arbitrage portfolio are all equal to zero.

(Module 36.1, LOS 36.a)

Marianne Belair, CFA, is a wealth manager for a well-known company in Paris, France. She has developed macroeconomic factor models on portfolios Alpha and Bravo.

Equations for the two portfolios:

$$R_{\text{Alpha}} = 0.08 - 0.7 F_{\text{INFL}} + 1.2 F_{\text{GDP}}$$

$$R_{\text{Bravo}} = 0.13 + 0.6 F_{\text{INFL}} + 2.3 F_{\text{GDP}}$$

Belair has asked her colleague Pierre Louboutin to calculate the return attributable to a 1.5% surprise in GDP for an equally weighted portfolio comprising Alpha and Bravo.

Meanwhile, Belair is looking at Merci, a beauty stock for which she has developed a macroeconomic factor model. The arbitrage-pricing model shows a required return of 10% and the company-specific surprise for the year was 2%. Exhibit 1 shows additional information on the model:

Exhibit 1:

Variable	Actual Value (%)	Expected Value (%)	Factor Sensitivity
Interest rates	3.5%	2.5%	−0.3
Unemployment level	6.5%	5.5%	−0.7

Emily Grant, a senior manager at the firm, asks Louboutin to analyze the performance of three managers using the information in Exhibit 2.

Exhibit 2: Decomposing Active Risk

Portfolio	Active factor risk squared (%)	Active specific risk squared (%)	Active risk squared (%)	Active factor risk (% of Total Active Risk)	Active specific risk (% of Total Active Risk)	Active risk (%)
EM	0.5	0.5	1	50	50	1
EC	25.2	10.8	36	70	30	6
EV	21.6	14.4	36	60	40	6

Finally, Belair would like to capitalize on her expectation that real business activity will increase over the next year. As a separate concern, she has some existing positive exposure to inflation risk, which she would like to hedge. To achieve her goals she can use the portfolios in the Exhibit 3 which show the five relevant factors and respective factor sensitivities:

Exhibit 3:

Risk Factor	A	B	C	D	E
Confidence	0.10	1.00	0.00	0.70	0.00

Time horizon 0.00 0.00 0.00 0.50 0.00

Inflation 1.00 0.00 0.00 0.30 1.00

Business cycle 0.90 1.00 1.00 0.00 0.00

Market timing 1.00 0.00 0.00 0.90 0.00

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Question ID: 1473957

Pierre's answer to Belair's first request regarding the equally weighted portfolio, is *closest* to:

A) 1.75%.



B) 2.13%.



C) 2.63%.



Explanation

The combined sensitivity of the GDP factor is $0.5 \times 1.2 + 0.5 \times 2.3 = 1.75$.

The change for 1.5% surprise is $1.5 \times 1.75 = 2.625\%$.

(Module 36.2, LOS 36.d)

Question #14 - 16 of 37

Question ID: 1473958

The actual return of Merci is *closest* to:

A) 9%.



B) 10%.



C) 11%.



Explanation

Key things to remember are to start with the APT required / expected return, then use surprises rather than actual values for each factor (i.e., actual minus expected), and finally to also add/subtract the company-specific surprise.

$$\text{actual return} = 10 - 0.3 \times (3.5 - 2.5) - 0.7 \times (6.5 - 5.5) + 2 = 11\%$$

(Module 36.2, LOS 36.d)

Question #15 - 16 of 37

Question ID: 1473959

Using Exhibit 2, the portfolio that has the *most* exposure to asset selection risk is:

A) EM.



B) EC.



C) EV.



Explanation

Asset selection risk is also known as active specific risk. Portfolio EV has the highest risk at 14.4.

(Module 36.3, LOS 36.f)

Question #16 - 16 of 37

Question ID: 1473960

Which two portfolios from Exhibit 3 best achieve Belair's goals in relation to business activity and inflation risk?

A) B and A.



B) B and E.



C) C and E.



Explanation

This question is about factor portfolios (i.e., a portfolio that has a sensitivity of one to a particular factor and zero to all other factors). The only two factor portfolios in Exhibit 3 are C and E having exposure to the business cycle and inflation respectively. A short position in portfolio E would eliminate the inflation risk.

(Module 36.3, LOS 36.f)

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Question ID: 1473941

A multi-factor model that uses unexpected changes (surprises) in macroeconomic variables (e.g., inflation and gross domestic product) as the factors to explain asset returns is called a:

- A) macroeconomic factor model.
- B) fundamental factor model.
- C) statistical factor model.



Explanation

Macroeconomic factor models use unexpected changes (surprises) in macroeconomic variables as the factors to explain asset returns. One example of a factor in this type of model is the unexpected change in gross domestic product (GDP) growth. In fundamental factor models, the factors are characteristics of the stock or the company that have been shown to affect asset returns, such as book-to-market or price-to-earnings ratios. A statistical factor model identifies the portfolios that best explain the historical cross-sectional returns or covariances among assets. The returns on these portfolios represent the factors.

(Module 36.2, LOS 36.d)

Question #18 of 37

Question ID: 1473961

A portfolio manager uses a two-factor model to manage her portfolio. The two factors are confidence risk and time-horizon risk. If she wants to bet on an unexpected increase in the confidence risk factor (which has a positive risk premium), but hedge away her exposure to time-horizon risk (which has a negative risk premium), she should create a portfolio with a sensitivity of:

- A) -1.0 to the confidence risk factor and 1.0 to the time-horizon factor.
- B) 1.0 to the confidence risk factor and -1.0 to the time-horizon factor.
- C) 1.0 to the confidence risk factor and 0.0 to the time-horizon factor.



Explanation

She wants to create a confidence risk factor portfolio, which has a sensitivity of 1.0 to the confidence risk factor and 0.0 to the time horizon factor. Because the risk premium on the confidence risk factor is positive, an unexpected increase in this factor will increase the returns on her portfolio. The exposure to the time-horizon risk factor has been hedged away, because the sensitivity to that factor is zero.

(Module 36.3, LOS 36.g)

Question #19 of 37

Question ID: 1473936

Michael Paul, a portfolio manager, is screening potential investments and suspects that an arbitrage opportunity may be available. The three portfolios that meet his screening criteria are detailed below:

Portfolio	Expected Return	Beta
X	12%	1.0
Y	16%	1.3
Z	8%	0.9

Which of the following portfolio combinations produces the highest return while maintaining a beta of 1.00?

	<u>Portfolio X</u>	<u>Portfolio Y</u>	<u>Portfolio Z</u>	
A)	25%	50%	25%	
B)	50%	12%	38%	
C)	100%	0%	0%	

Explanation

Portfolio Weights			Expected Return	Beta
X	Y	Z		
25%	50%	25%	13.00%	1.13
50%	12%	38%	10.96%	1.00
100%	0%	0%	12.00%	1.00

Portfolio weights of 25%, 50%, and 25% yield the highest return, but at a beta of 1.13. Investing 100% in Portfolio X yields the highest return for this risk level (i.e., beta = 1.00).

(Module 36.1, LOS 36.b)

Question #20 of 37

Question ID: 1473940

Portfolios A and B have an expected return of 4.4% and 5.3% respectively. Assume that a one-factor APT model is appropriate and the factor sensitivities of portfolios A and B are 0.8 and 1.1 respectively. The risk-free rate and factor risk premium are *closest* to:

	<u>Risk Free Rate</u>	<u>Factor Risk Premium</u>	
A)	3.00%	2.00%	
B)	2.50%	3.00%	
C)	2.00%	3.00%	

Explanation

Expected return = risk free rate + factor sensitivity x risk premium

For portfolio A: $0.044 = R_f + 0.8\lambda$ Hence $R_f = 0.044 - 0.8\lambda$

Substituting $R_f = (0.04 - 0.8\lambda)$ for portfolio B, $0.053 = (0.044 - 0.8\lambda) + 1.1\lambda$

$\lambda = 0.03$ or 3% and $R_f = 2\%$.

(Module 36.1, LOS 36.c)

Question #21 of 37

Question ID: 1473951

A common strategy in bond portfolio management is *enhanced indexing by matching primary risk factors*. This strategy could be implemented by forming:

- A) a portfolio with factor sensitivities that sum to one. 
- B) a portfolio with factor sensitivities equal to that of the index. 
- C) a portfolio with asset portfolio weights equal to that of the index. 

Explanation

Enhanced indexing by matching primary risk factors could be implemented by creating a tracking portfolio with the same factor sensitivities as the index but with a different set of bonds. Then any differences in performance between the portfolio and the benchmark index will be the result of bond selection ability and not from different exposures to macroeconomic factors like GDP, inflation, and interest rates.

(Module 36.3, LOS 36.f)

Question #22 of 37

Question ID: 1473944

Summer Vista decides to develop a fundamental factor model. She establishes a proxy for the market portfolio, and then considers the importance of various factors in determining stock returns. She decides to use the following factors in her model:

- Changes in payout ratios.
- Credit rating changes.
- Companies' position in the business cycle.
- Management tenure and qualifications.

Which of the following factors is *least appropriate* for Vista's factor model?

- A) Management tenure and qualifications. 
- B) Changes in payout ratios. 
- C) Companies' position in the business cycle. 

Explanation

Fundamental factors are factors measured by characteristics of the companies themselves, like price-to-earnings (P/E) ratios or growth rates. Macroeconomic factors are economic influences on security returns. A company's position in the business cycle is dependent on the cycle itself, and cannot be accurately measured by looking at a company's fundamentals – business cycle is a macroeconomic factor. Payout ratios and management tenure are pieces of company-specific data suitable for use in a fundamental factor model.

(Module 36.2, LOS 36.d)

Question #23 of 37

Question ID: 1473949

A portfolio with a factor sensitivity of one to a particular factor in a multi-factor model and zero to all other factors is called a(n):

A) tracking portfolio.



B) factor portfolio.



C) arbitrage portfolio.

**Explanation**

A factor portfolio is a portfolio with a factor sensitivity of one to a particular factor and zero to all other factors. An arbitrage portfolio is a portfolio with factor sensitivities of zero to all factors, positive expected net cash flow, and an initial investment of zero. A tracking portfolio is a portfolio with a specific set of factor sensitivities designed to replicate the factor exposures of a benchmark index.

(Module 36.3, LOS 36.f)

Question #24 of 37

Question ID: 1473934

Arbitrage pricing models assume which risk is priced?

A) Systematic.



B) Both systematic and unsystematic.



C) Unsystematic.

**Explanation**

Unsystematic risk can be diversified away. Thus, arbitrage pricing reflects only systematic risk. It is assumed that the portfolio manager will take steps to diversify and reduce risk.

(Module 36.1, LOS 36.a)

Question #25 of 37

Question ID: 1473963

Janice Barefoot, CFA, has managed a portfolio where she used the Dow Jones Industrial Average (DJIA) as a benchmark. In the past two years the average monthly return on her portfolio has been higher than that of the DJIA. To get a measure of active return per unit of active risk Barefoot should compute the:

- A)** Sharpe ratio, which is the standard deviation of the differences between the portfolio and benchmark returns divided into the average of those differences. 
- B)** information ratio, which is the average excess portfolio return over the benchmark divided by the standard deviation of the differences between the portfolio and benchmark returns. 
- C)** information ratio, which is the standard deviation of the differences between the portfolio and benchmark returns divided by the average of those differences. 

Explanation

The information ratio is the measure of active return per unit of active risk. If we let $X = (\text{monthly portfolio return} - \text{the benchmark return})$, then the information ratio = (the average of X / the standard deviation of X). It is similar to the Sharpe ratio, which defines the random variable Y as $Y = (\text{monthly portfolio return} - \text{the risk-free rate})$. The Sharpe ratio = (the average of Y / the standard deviation of the portfolio return) = the standard deviation of Y if the risk-free rate is constant.

(Module 36.3, LOS 36.e)

Question #26 of 37

Question ID: 1473942

The macroeconomic factor models for the returns on Omni, Inc., (OM) and Garbo Manufacturing (GAR) are:

$$R_{OM} = 20.0\% + 1.0(F_{GDP}) + 1.4(F_{QS}) + \varepsilon_{OM}$$

$$R_{GAR} = 15.0\% + 0.5(F_{GDP}) + 0.8(F_{QS}) + \varepsilon_{GAR}$$

What is the expected return on a portfolio invested 60% in Omni and 40% in Garbo?

- A)** 18.0%. 
- B)** 20.96%. 
- C)** 19.96%. 

Explanation

In the macroeconomic model, the intercept is the expected return. The expected return of the portfolio is the weighted average of the expected return of the 2 stocks:

$$R_p = [(0.6)(20.0\%) + (0.4)(15.0\%)] = 18\%$$

(Module 36.2, LOS 36.d)

Question #27 of 37

Question ID: 1473927

Which of the following is an equilibrium-pricing model?

A) The arbitrage pricing theory (APT).



B) Fundamental factor model.



C) Macroeconomic factor model.



Explanation

The APT is an equilibrium-pricing model; multi-factor models are "ad-hoc," meaning the factors in these models are not derived directly from an equilibrium theory. Rather they are identified empirically by looking for macroeconomic variables that best fit the data.

(Module 36.1, LOS 36.a)

Question #28 of 37

Question ID: 1473937

Given a three-factor arbitrage pricing theory (APT) model, what is the expected return on the Premium Dividend Yield Fund?

- The factor risk premiums to factors 1, 2 and 3 are 8%, 12% and 5%, respectively.
- The fund has sensitivities to the factors 1, 2, and 3 of 2.0, 1.0 and 1.0, respectively.
- The risk-free rate is 3.0%.

A) 33.0%.



B) 36.0%.



C) 50.0%.



Explanation

The expected return on the Premium Dividend Yield Fund is $3\% + (8.0\%)(2.0) + (12.0\%)(1.0) + (5.0\%)(1.0) = 36.0\%$.

(Module 36.1, LOS 36.c)

Question #29 of 37

Question ID: 1473954

Janice Barefoot, CFA, has been managing a portfolio for a client who has asked Barefoot to use the Dow Jones Industrial Average (DJIA) as a benchmark. In her second year, Barefoot used 29 of the 30 DJIA stocks. She selected a non-DJIA stock in the same industry as the omitted DJIA stock to replace that stock. Compared to the DJIA, Barefoot placed a lower weight on the communication stocks and a higher weight on the other stocks still in the portfolio. Over that year, the non-DJIA stock in the portfolio had a positive and higher return than the omitted DJIA stock. The communication stocks had a negative return while all of the other stocks had a positive return. The portfolio managed by Barefoot outperformed the DJIA. Based on this we can say that the return from factor tilts and asset selection were:

- A) both positive. 
- B) negative and positive respectively. 
- C) positive and negative respectively. 

Explanation

Since the communications stocks had a negative return while all the other stocks had a positive return, Barefoot's underweighting of those stocks produced a positive tilt return. Since the asset chosen to replace the DJIA stock outperformed the omitted stock, the asset selection return was positive.

(Module 36.3, LOS 36.f)

Question #30 of 37

Question ID: 1473955

Rob Tanner, portfolio manager at Alpha Inc. meets his old college friend Del Torres for lunch. Torres excitedly tells Tanner about his latest work with tracking and factor portfolios. Torres says he has developed a tracking portfolio to aid in speculating on oil prices and is working on a factor portfolio with a specific set of factor sensitivities to the Russell 2000.

Did Torres correctly describe tracking and factor portfolios?

- | | <u>Tracking</u> | <u>Factor</u> | |
|--------|-----------------|---------------|---|
| A) Yes | No | |  |
| B) No | Yes | |  |

C) No No



Explanation

Torres reversed the concepts and is thus incorrect on both counts. A factor portfolio is a portfolio with a factor sensitivity of 1 to a particular factor and zero to all other factors. It represents a pure bet on one factor, and can be used for speculation or hedging purposes. A tracking portfolio is a portfolio with a specific set of factor sensitivities. Tracking portfolios are often designed to replicate the factor exposures of a benchmark index like the Russell 2000.

(Module 36.3, LOS 36.f)

Question #31 of 37

Question ID: 1473938

Given a three-factor arbitrage pricing theory APT model, what is the expected return on the Freedom Fund?

- The factor risk premiums to factors 1, 2, and 3 are 10%, 7% and 6%, respectively.
- The Freedom Fund has sensitivities to the factors 1, 2, and 3 of 1.0, 2.0 and 0.0, respectively.
- The risk-free rate is 6.0%.

A) 24.0%.



B) 30.0%.



C) 33.0%.



Explanation

The expected return on the Freedom Fund is $6\% + (10.0\%)(1.0) + (7.0\%)(2.0) + (6.0\%)(0.0) = 30.0\%$.

(Module 36.1, LOS 36.c)

Question #32 of 37

Question ID: 1473962

In the context of multi-factor models, investors with lower-than-average exposure to recession risk (e.g. those without labor income) can earn a risk premium for holding dimensions of risk unrelated to market movements by creating equity portfolios with:

A) greater-than-average market risk exposure.



B) greater-than-average exposure to the recession risk factor.



C) less-than-average exposure to the recession risk factor.



Explanation

Multifactor models allow us to capture other dimensions of risk besides overall market risk. Investors with unique circumstances different than the average investor may want to hold portfolios tilted away from the market portfolio in order to hedge or speculate on factors like recession risk, interest rate risk or inflation risk. An investor with lower-than-average exposure to recession risk can earn a premium by creating greater-than-average exposure to the recession risk factor. In effect, he earns a risk premium determined by the average investor by taking on a risk he doesn't care about as much as the average investor does.

(Module 36.3, LOS 36.g)

Question #33 of 37

Question ID: 1473946

Assume you are considering forming a common stock portfolio consisting of 25% Stonebrook Corporation (Stone) and 75% Rockway Corporation (Rock). As expressed in the two-factor returns models presented below, both of these stocks' returns are affected by two common factors: surprises in interest rates and surprises in the unemployment rate.

$$R_{\text{Stone}} = 0.11 + 1.0F_{\text{Int}} + 1.2F_{\text{Un}} + \epsilon_{\text{Stone}}$$

$$R_{\text{Rock}} = 0.13 + 0.8F_{\text{Int}} + 3.5F_{\text{Un}} + \epsilon_{\text{Rock}}$$

Assume that at the beginning of the year, interest rates were expected to be 5.1% and unemployment was expected to be 6.8%. Further, assume that at the end of the year, interest rates were actually 5.3%, the actual unemployment rate was 7.2%, and there were no company-specific surprises in returns. This information is summarized in Table 1 below:

Table 1: Expected versus Actual Interest Rates and Unemployment Rates

	Actual	Expected	Company-specific returns surprises
Interest Rate	0.053	0.051	0.0
Unemployment Rate	0.072	0.068	0.0

What is the portfolio's sensitivity to interest rate surprises?

A) 0.25.



B) 0.85.



C) 0.95.



Explanation

The portfolio composition is 25% Stonebrook and 75% Rockway. The interest rate sensitivities for Stonebrook and Rockway are 1.0 and 0.8, respectively. Thus, the portfolio's sensitivity to interest rate surprises is: $(0.25)(1.0) + (0.75)(0.8) = 0.85$.

(Module 36.2, LOS 36.d)

Question #34 of 37

Question ID: 1473952

The Real Value Fund is designed to have zero exposure to inflation. However its current inflation factor sensitivity is 0.30. To correct for this, the portfolio manager should take a:

- A) 30% short position in the inflation tracking portfolio.
- B) 30% long position in the inflation factor portfolio.
- C) 30% short position in the inflation factor portfolio.



Explanation

To hedge inflation, the fund should take a 30% short position in the inflation factor portfolio. This short position will fully offset the fund's positive exposure to inflation. Tracking portfolios are typically used for active asset selection and have multiple factor exposures which would prevent them from adequately hedging the inflation exposure of the fund.

(Module 36.3, LOS 36.f)

Question #35 of 37

Question ID: 1473929

Which of the following is NOT an underlying assumption of the arbitrage pricing theory (APT)?

- A) A market portfolio exists that contains all risky assets and is mean-variance efficient.
- B) Asset returns are described by a K factor model.
- C) There are a sufficient number of assets for investors to create diversified portfolios in which firm-specific risk is eliminated.



Explanation

The APT makes no assumption about a market portfolio.

(Module 36.1, LOS 36.a)

Question #36 of 37

Question ID: 1473945

Assume you are considering forming a common stock portfolio consisting of 25% Stonebrook Corporation (Stone) and 75% Rockway Corporation (Rock). As expressed in the two-factor returns models presented below, both of these stocks' returns are affected by two common factors: surprises in interest rates and surprises in the unemployment rate.

$$R_{\text{Stone}} = 0.11 + 1.0F_{\text{Int}} + 1.2F_{\text{Un}} + \epsilon_{\text{Stone}}$$

$$R_{\text{Rock}} = 0.13 + 0.8F_{\text{Int}} + 3.5F_{\text{Un}} + \epsilon_{\text{Rock}}$$

Assume that at the beginning of the year, interest rates were expected to be 5.1% and unemployment was expected to be 6.8%. Further, assume that at the end of the year, interest rates were actually 5.3%, the actual unemployment rate was 7.2%, and there were no company-specific surprises in returns. This information is summarized in Table 1 below:

Table 1: Expected versus Actual Interest Rates and Unemployment Rates

	Actual	Expected	Company-specific returns surprises
Interest Rate	0.053	0.051	0.0
Unemployment Rate	0.072	0.068	0.0

What is the expected return for Stonebrook in the absence of surprises?

- A) 11.0%.
- B) 13.2%.
- C) 13.0%.

**Explanation**

The expected return for Stonebrook is simply the intercept return (a_i) of 0.11, or = 11.0%.

(Module 36.2, LOS 36.d)

Question #37 of 37

Question ID: 1473948

A portfolio with a specific set of factor sensitivities designed to replicate the factor exposures of a benchmark index is called a:

A) arbitrage portfolio.



B) tracking portfolio.



C) factor portfolio.



Explanation

A tracking portfolio is a portfolio with a specific set of factor sensitivities designed to replicate the factor exposures of a benchmark index. A factor portfolio is a portfolio with a factor sensitivity of one to a particular factor and zero to all other factors. An arbitrage portfolio is a portfolio with factor sensitivities of zero to all factors, positive expected net cash flow, and an initial investment of zero.

(Module 36.3, LOS 36.f)